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The Clip Moves, the Note Doesn't: An Interrupted Time-Series Analysis of a Classroom Behaviour-Chart Threshold Reset

Renke Sabel, Osei Vandermeer

School of Continuous Improvement, Slop University

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Abstract. A primary-school clip chart moves a child's name up or down a coloured ladder in response to logged behaviour, and a digitised, network-wide version of the same instrument logs every movement automatically. This paper reports an interrupted time-series analysis of a scheduled, network-wide recalibration that lowered the clip chart's top-tier threshold from five clip-ups to four, delivered identically and on the same day to 26 classrooms across nine primary schools. Combining fourteen weeks of digitised clip-movement logs with the schools' independently kept incident-note record — the notes sent home documenting a behaviour concern — for 612 children, we fit a segmented regression around the reset date. Weekly clip-up movements per child show an immediate, sustained level shift of +1.3 (95% CI 1.05-1.55, $p < 0.001$) following the recalibration; the incident-note rate the chart notionally exists to reduce shows no corresponding shift (-0.02, 95% CI −0.09 to 0.05, n.s.). A classroom-level comparison of the two changes finds no relationship between how far a classroom's clip rate moved and whether its note rate moved at all ($r = 0.09$, $n = 26$). An ablation restricting the sample to classrooms led by a first-year teacher reproduces the same decoupling (n.s. difference from the pooled estimate); we retain the pooled sample for completeness. The chart, on this evidence, tracked the reset far more faithfully than it tracked the behaviour it was installed to record.

Introduction

A clip chart is furniture in most primary classrooms before it is anything else: a laminated ladder pinned near the whiteboard, a peg or magnet bearing each child's name, moved up a tier for behaviour a teacher wants to encourage and down for behaviour a teacher wants to discourage. Digitised versions of the same instrument now log every movement automatically, timestamped and exportable, and — increasingly — configurable at the platform level rather than the classroom level: a vendor can change how many clip-ups a tier requires for every participating classroom in a single update.

That configurability is what makes the clip chart, ordinarily an instrument that resists controlled study, briefly tractable. A behaviour-tracking platform used across a regional network of primary schools scheduled a routine tier recalibration partway through a term, lowering the number of clip-ups required to reach the top ("Outstanding") tier from five to four. The change was pushed to all 26 participating classrooms on the same day as part of the platform's own end-of-term defaults refresh, with no classroom opting in, opting out, or receiving it early — a uniform, dated, exogenous shift in the instrument's calibration, sitting alongside an independent record the instrument does not touch: the incident notes each school's front office sends home when a behaviour concern is significant enough to document for a parent.

Whether an instrument's readings and the record it is meant to track move together under a rule change neither party requested is, to our knowledge, untested for the classroom clip chart specifically, though the broader question

— whether a measurement device and the thing it measures can decouple even under a uniform, well-intentioned change — recurs across institutional measurement generally. This paper reports an interrupted time-series analysis of the recalibration's effect on both series: fourteen weeks of digitised clip-movement logs and the schools' own incident-note record, for 612 children across the 26 classrooms. The paper's contributions:

1. We assemble what is, to our knowledge, the first matched clip-movement and incident-note dataset spanning a mid-deployment behaviour-instrument recalibration in primary classrooms.
2. Using a segmented-regression interrupted time series, we find the recalibration produced an immediate, sustained level shift in logged clip movements, while the incident-note record showed no corresponding shift.
3. We show the decoupling holds when the sample is restricted to classrooms led by a first-year teacher, suggesting the effect does not depend primarily on implementation experience.

None of this suggests any teacher, school, or platform acted in bad faith; the recalibration was a routine defaults update, applied identically everywhere. It suggests only that an instrument's own reading and the behaviour it is meant to track need not move together, even when the instrument changes for every classroom on the same day, for the same stated reason.

Related work

Digitised behaviour-tracking platforms extend the clip chart's basic logic — a visible, incrementable score attached to a child — into a permanent, exportable record, and the resulting datafication of classroom discipline has drawn direct critique for shifting a teacher's attention toward the instrument's own display [1]. The wider token-economy literature the clip chart descends from is not univocal: a systematic evaluation applying What Works Clearinghouse single-case-design standards could not classify token economies as evidence-based [2], while a larger effect-size meta-analysis across K-5 settings found large effects in the great majority of studies alongside procedural documentation too vague, in many cases, to support replication [3], and an early classroom-participation application found the token itself, not only the reinforced behaviour, became what students tracked [4]. The broader school-wide behaviour-support framework these platforms increasingly sit inside is separately reported to reduce exclusionary-discipline outcomes for students with disabilities [5], a school-level effect this paper's individual-classroom design cannot speak to either way.

That an instrument can quietly stop measuring the thing it was built for is a recurring finding once a measure enters routine institutional use: audit regimes reshape the practice they monitor [6], publication metrics distort the publishing behaviour that produces them once authors optimise directly against the count [7], and public-sector performance targets generate documented gaming even under close administrative attention [8] — a pattern prior Slop University work traced in an internal indicator registry, where a sunset scorer's own uptime outlasted the great majority of indicators it flagged for retirement [9]. A parallel literature on stated versus enacted measures finds self-report a weak proxy for logged behaviour generally [10], [11], a gap this paper's own institution has also logged at the checkout counter [12]. Interrupted time series via segmented regression is the standard design for a dated, uniform intervention where randomisation is unavailable [13], [14], with regression discontinuity the nearest sibling design where assignment follows a running variable rather than a date [15]. Teacher experience separately predicts classroom-management practice generally [16], [17], motivating this paper's ablation.

Method

Setting and platform

Data were drawn from a shared digitised clip-chart platform used across 26 classrooms in nine public primary schools within one regional network. Each classroom's physical clip chart is paired with a classroom tablet on which the teacher, or in most observed classrooms a rostered student "chart monitor," logs each clip movement in real time; the platform aggregates movements into a running weekly count per child. Front-office incident notes — the record sent home to a parent when a behaviour concern is judged significant enough to document — are logged independently in each school's own administrative system and were not visible to whoever was logging clip movements that week.

The recalibration

Partway through the term, the platform vendor pushed a scheduled configuration update lowering the "Outstanding" tier's threshold from five net clip-ups to four, delivered identically to all 26 classrooms on the same calendar day as part of the platform's routine end-of-term defaults refresh. No classroom opted in, opted out, or received the update on a different day; the change was announced to schools only as a minor platform update, not as a behaviour-management intervention, and no accompanying guidance on expected effects was distributed.

Sample and window

Schools	9
Classrooms	26
Children	612
Weeks (pre / post)	7 / 7
Child-weeks per series	8,568

The analysis draws on 612 children across the 26 classrooms, spanning fourteen weeks: seven immediately preceding the recalibration and seven immediately following.

Interrupted time-series model

For each series — weekly clip-up movements per child; weekly incident notes per child — we fit a segmented regression

$$Y_t = \beta_0 + \beta_1 T_t + \beta_2 D_t + \beta_3 D_t (T_t - t_0) + \varepsilon_t,$$

where T_t indexes the study week, D_t is an indicator equal to 1 from the recalibration date onward and 0 before it, and t_0 is the week of the recalibration. β_2 estimates the immediate level shift at the reset; β_3 estimates any accompanying change in weekly trend. Standard errors are clustered by classroom to account for repeated child-weeks logged under the same clip chart.

Ablation

Classrooms were separately coded by lead-teacher experience — first-year teacher (7 classrooms) versus a teacher with two or more years in the role (19 classrooms) — from each school's own staffing record, to test whether the recalibration's effect on clip movements depends on how practised the classroom's teacher is with the platform generally.

Results

Across the fourteen-week window, weekly clip-up movements per child hold close to a flat pre-reset average before stepping up sharply the week the recalibration took effect, while the incident-note rate does not move.

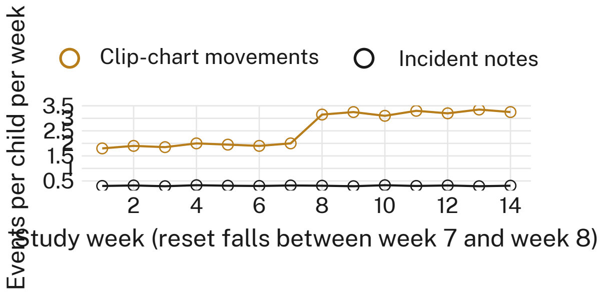

Figure 1: Weekly clip-chart movements per child step up immediately after the reset (week 7 to week 8); the weekly incident-note rate, on the same per-child-per-week scale, does not.

The segmented regression in Figure 1 confirms the step is a level shift, not a trend change: the clip-movement series shows an immediate shift of +1.3 movements per child per week (95% CI 1.05-1.55, $p < 0.001$) with no significant change in the pre-or post-reset slope (both n.s.), while the incident-note series shows a level-shift estimate of −0.02 (95% CI −0.09 to 0.05, n.s.) and likewise no slope change.

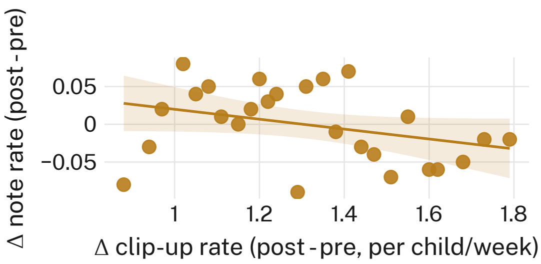

Figure 2: Per-classroom change in weekly clip-up rate against the matching change in weekly incident-note rate (n = 26 classrooms). The fitted line is close to flat (r = 0.09): how far a classroom's clip rate moved after the reset carries no information about whether its note rate moved at all.

Figure 2 rules out the possibility that the aggregate null in the note series is masking offsetting classroom-level effects: classrooms whose clip-up rate rose most sharply were no more or less likely to see their incident-note rate rise, fall, or hold steady than classrooms whose clip-up rate rose only modestly.

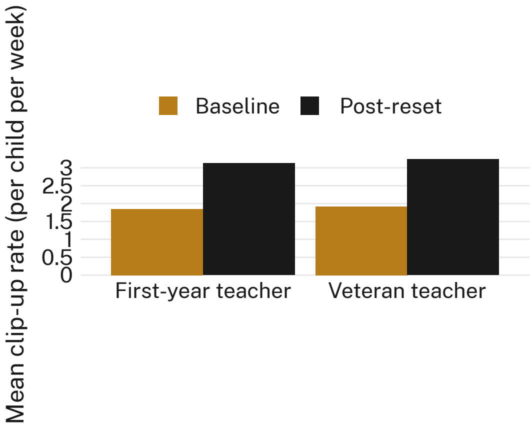

Figure 3: Mean weekly clip-up rate, baseline versus post-reset, by lead-teacher experience. The level shift is close to identical for first-year (7 classrooms) and veteran (19 classrooms) teachers.

The ablation in Figure 3 changes almost nothing. The estimated level shift is 1.28 (95% CI 0.86-1.70) for first-year-teacher classrooms and 1.32 (95% CI 1.02-1.62) for veteran-teacher classrooms, a difference of 0.04 that a two-sample comparison does not distinguish from noise ($p = 0.81$). We retain the pooled estimate in the headline result for completeness, and because teacher experience was logged at the classroom rather than the child level and cannot itself explain a within-classroom decoupling.

Discussion

Two independent measures point the same way. A network-wide, dated, identically delivered recalibration of a behaviour-tracking instrument's threshold produced an immediate, sustained shift in what the instrument logged, and no detectable shift in the record the instrument exists to move — not in aggregate, not classroom by classroom, and not once the sample is narrowed to classrooms led by a first-year teacher, where an implementation effect might plausibly have shown up first. Neither result shows the clip chart does nothing; a chart that structures daily attention around a visible, incrementable score is doing something, and this design cannot rule out effects the incident-note record would not capture. It suggests only that, for the specific claim a threshold recalibration would need to support — that an easier-to-reach top tier changes documented behaviour, not only logged clip positions — this dataset offers no support.

The pattern sits comfortably alongside the University's own prior finding that an indicator's uptime can outlast the great majority of what it was built to flag [9]: an instrument continuing to register cleanly is not evidence that the thing behind it moved. Nothing here suggests the platform vendor, the schools, or any teacher engineered the outcome; the recalibration was announced, and received, as routine software maintenance. It suggests only that a measurement device and the phenomenon it measures are not guaranteed to move together, even under a change applied uniformly, on one day, for the same stated reason, everywhere at once.

Limitations

The recalibration's uniform, same-day delivery is this design's strength and its constraint: with no staggered or held-back comparison group, the interrupted time series cannot rule out an unrelated, coincident event affecting all 26 classrooms in the same week, though no school reported one and the incident-note series' own flat trend through the reset week argues against a shared shock large enough to move behaviour. The fourteen-week window is short enough that a slower-building effect on documented behaviour remains possible; the consistency of the null across both the aggregate and classroom-level analyses suggests any such effect is not large, though it does not rule one out. Incident-note issuance itself reflects front-office and teacher judgement about what merits documentation, a threshold this design assumes but did not audit directly.

Conclusion

A behaviour-chart recalibration delivered identically to 26 classrooms on the same day produced an immediate, sustained shift in what the chart logged, and no shift, in aggregate or classroom by classroom, in the incident-note record the chart is meant to reduce. The instrument, on this evidence, tracked its own reset more faithfully than it tracked the behaviour it was installed to change. Future work might extend the design to a second recalibration at the same schools, to test whether the decoupling itself is stable across repeated resets, and to the front-office judgement behind an incident note directly, which this design borrowed as ground truth but did not examine on its own terms.

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